A REDUCED LOCALLY ALGEBRAIC GROUP SCHEME OVER A PERFECT FIELD OF CHARACTERISTIC P IS SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-group-scheme-module-differentials (canonical tag 047I). cited at proof lines 1523-1525. The sheaf of differentials Omega_{G/k} is free.
- Lemma varieties-lemma-char-p-differentials-free-smooth (canonical tag 04QP). cited at proof lines 1525-1526. Perfect field, reduced, free differentials imply smoothness.
- Lemma lemma-group-scheme-characteristic-zero-smooth (canonical tag 047N). cited at statement lines 1513-1515. Cross-reference to the characteristic-zero companion statement.

Source reference. Original lemma 047N: sources/stacks-chow/tags.txt, source lines 5514–5514. Cited at statement lines 1513-1515 (parenthetical cross-reference to the characteristic-zero case); contextual reference.

Source reference. Original lemma 047I: sources/stacks-chow/tags.txt, source lines 5509–5509. Cited at proof lines 1523-1525: Omega_{G/k} is free; load-bearing.

Source reference. Original lemma 04QP: sources/stacks-chow/tags.txt, source lines 6157–6157. Cited at proof lines 1525-1526: free differentials plus reducedness over a perfect field give smoothness; load-bearing.

Lemma 0.2. *Let k be a perfect field of characteristic $p > 0$ (see Lemma 047N for the characteristic zero case). Let G be a locally algebraic group scheme over k. If G is reduced then the structure morphism $G \to \mathrm{Spec}(k)$ is smooth, i.e., G is a smooth group scheme.*

Proof. By Lemma 047I the sheaf $\Omega_{G/k}$ is free. Hence the lemma follows from Varieties, Lemma 04QP. □

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.